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Predicting Choke Valve Erosion in HPHT Gas Wells Using Machine Learning

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Abstract

Production choke valves are essential to control pressure and flow rate of the produced hydrocarbon stream from a well. These choke valves are subject to continuous erosion that results from the interaction of particles in the hydrocarbon fluids. Traditionally, detection of choke erosion in high-pressure high-temperature (HPHT) gas wells is done through physical inspection of the production choke to confirm erosion. This process is highly time consuming and expensive due to the need to interrupt production, depressurize the flowline, decoupling the choke body from the flowline, and performing choke body inspection on site or at the workshop, if erosion is confirmed then a choke replacement is scheduled. In this paper, we present a revolutionary approach that utilizes machine learning techniques to predict the remaining useful life (RUL) for chokes in gas wells. Our findings offer a multitude of business values such as maximized equipment efficiency, cost reduction, enhanced safety, and improved production. The solution utilizes an advanced regression algorithm, incorporating real-time well performance variables. These inputs communally contribute to predict RUL for multiple HPHT gas wells. Furthermore, thorough data discovery, transformation, and quality control procedures, our predictive model can predict the days to erosion based on current well status. The work provided strong evidence for the effectiveness of this AI-driven approach, with an accuracy exceeding 95% in predicting the RUL for gas wells. The combination of established digital dashboard powered with this AI-predicted RUL enables engineers to quickly identify and prioritize wells susceptible to erosion. Thus, enhancing surveillance to ensure well integrity & production efficiency. Application of this methodology also allowed for careful allocation of limited resources toward wells exhibiting erosion vulnerability, effectively reducing the cost, risk and downtime associated with such failures. By utilizing this predictive maintenance approach to predict RUL for chokes, operators can mitigate future asset failures, improve safety standards and reduce downtime.

Introduction

The most crucial part of well operations in oil and gas industry is to maintain well integrity at all times. The choke valves are crucial for regulating flow rates in all well types. However, high solids content in gas wells can potentially lead to choke erosion, resulting in production downtime and safety hazards. To address this issue, our solution predicts the time a choke is likely to wear out in a HPHT gas well. This

allows production engineers to plan ahead and perform maintenance before the failure occurs. Typically, detection of choke erosion in HPHT gas wells requires shutting down operations, depressurize the flowline, and shipping the choke to inspection facilities for inspection – a time-consuming and very costly process. However, our approach utilizes machine learning techniques to predict the RUL which will help to identify potential erosion before it becomes a major problem in the field. This enables scheduling maintenance during routine shutdowns, reducing unexpected interruptions and improving overall efficiency. By adopting this predictive technology, operators now can achieve significant gains in productivity, reduce expenses, enhance workplace safety, and increase production.

One of the major challenges for production engineers operating HPHT gas platforms is monitoring performance of their wells while resolving any well integrity issues that might occur. Particularly choke valves as they are a critical component utilized in surface controller mechanism for regulating flowing gas to the associated stages of production while also used to control pressure, optimize production and ensure safety of surface equipment. Chokes are of two types: dynamic or fixed, enabling operators to control the flow accurately based on the well conditions. With time, those chokes are exposed to wear or malfunction due to continuous exposure to high velocity gas. To avoid such cases, well integrity has to be maintained through a multi-layered envelope system. Overtime, choke valves become more vulnerable to degradation, requiring increased monitoring and frequent assessment to ensure their integrity. Environments such as high-pressure and high velocity which are usually found with hydrocarbon production in conjunction with corrosive properties of the fluids are the main contributor to erosion. Furthermore, port dimension has been proven to be affected by the erosion caused when particles with vast sizes erode the direct exposed choke surface. Understanding the causes of erosion is essential for developing effective monitoring and mitigation strategies. By utilizing and adopting the predictive maintenance approach for detecting early signs of choke erosion, production engineers will be able to achieve significant gains in maximizing equipment efficiency, enhance operations and workplace safety, increase assets lifespan, cost reduction, avoid unplanned production interruption, improve production planning and finally reducing the impact of pressure fluctuations on the of formation. [Sliem et al. \(2021\)](#)

Numerous studies have been accomplished showcasing breakthrough for detecting production choke defects in HPHT gas wells. Similarly, [Al-Kadem et al. \(2024\)](#) explored the application of real-time advisory tool to predict choke valve erosion in Offshore HPHT gas wells, by using well digital twin for simulation and analysis. To do so, they explored multiple parameters such as flow rate and ratios, pressure, temperature and actual choke settings as inputs to the digital model which is then connected to Microsoft Excel using tools such as Visual Basic macros and OpenServer. The integration of artificial intelligent (AI) with real-time erosion detection approaches, has been proven to be very effective in the oil and gas industry.

Moreover [Alghamdi et al. \(2021\)](#), demonstrated the usefulness of AI in detecting gas choke defects achieving above 0.9 correlation while accurately predicting the exact point where the choke performance began showing signs of abnormality. This AI tool has the power to assist production engineers to enhance choke performance while aligning with production strategy and compliance. This approach underscores the potential of AI-driven technologies in revolutionizing erosion prediction and management in HPHT gas wells. The advantages of implementing AI in detecting erosion within gas field are numerous. One of the main benefits are safety of both the well operators and the environments surrounding the fields, early detection of any abnormal activities of choke valves especially in HPHT gas wells can prevent failures and ensure the safe operation of wells. Other advantages of utilizing AI-driven models are minimizing operational cost by avoiding production interruption. Nevertheless, AI can improve decision-making by enabling production engineers to make informed decision by using AI as an advisory tool that can uncover hidden insights from real-time field data with high accuracy.

Leveraging AI in choke erosion domain poses significant hurdles, primarily due to the scarcity of high-quality data necessary to effectively train the model. Erosion data is inherently complex, noisy and highly influenced by various elements such as environmental factors, operational procedures and material

characteristics. To implement trustworthy models, it's essential to make sure the precision, accuracy and availability of this data. Additionally, to interpret the results of these models it requires a specialized knowledge in both production engineering and data science, which can pose a significant obstacle for some organizations to effectively bridge and implement these models.

Although significant hurdles exist, the potential advantages of leveraging AI in choke erosion management are significant. By accurately predicting choke erosion in HPHT, production engineers can now optimize maintenance scheduling, minimize downtime, and prolong the operational life of wells. Furthermore, AI's capability to uncover hidden patterns and trends within erosion data can reveal novel insights, enabling the development of more effective erosion management strategies that surpass traditional analytical approaches.

Problem Statement and Objective

Traditionally, detection of choke erosion in HPHT gas wells is done through physical inspection of the production choke to confirm erosion. This process is highly time consuming and expensive due to the need to interrupt production, depressurize the flowline, decouple the choke body from the flowline, and ship the choke body to inspection facilities for inspection, if erosion is confirmed then a choke replacement is scheduled.

Production engineers are tasked with the critical responsibility of continuously monitoring the performance of each well to ensure meeting production demand and identify any potential inconsistencies in well parameters that may disrupt the production, including rate & choke valve performance. However, this manual process is prone to human error, as engineers may misinterpret choke data and incorrectly predict erosion trends, potentially mistaking normal fluctuations for erosion, which can lead to inaccurate assessments and suboptimal decision-making.

This paper presents a revolutionary approach that utilizes machine learning techniques to predict the remaining useful life for chokes in gas wells.

Methodological Framework

Data Sources and Variables

The predictive maintenance framework implemented in this project was based on structured historical dataset for five different HPHT offshore gas wells. The dataset primarily included wells that had undergone choke replacement due to erosion in the past, providing a baseline for model training. The dataset was also split further for validation and testing, although the number of failure records per well varied significantly — from once every year to once every few years.

To fully understand erosion detection, the feature set used in this study covers multiple aspect of choke operating conditions. The main features incorporated into the model include:

Table 1—Feature type and name

Feature type	Feature name
Number	Upstream flowing wellhead pressure
Number	Downstream flowing wellhead pressure
Number	Upstream flowing wellhead temperature
Number	Downstream flowing wellhead temperature
Number	Rate
Number	Choke size opening
Date	Installation date

Feature type	Feature name
Date	Choke replacement date

The initial phase for data exploration included assessment of feature availability, quality and their correlation to erosion risk. During this phase, we excluded multiple features due to different reasons such as their weak or non-existent relationship in regards to signs of erosion or improper data quality. Although the dataset used in this study has certain limitation, its overall structure had been enhanced by utilizing different feature engineering methods and logics during the preprocessing phase, allowing the model to predict hidden insights and patterns related to RUL of chokes.

Data Preparation Process

The development of our predictive model required a rigorous dataset preprocessing, which consisted of several crucial steps as illustrated in Figure 1, aimed at preserving data quality, eliminating irrelevant information, and optimizing the model's capacity for generalization. Given the complexity of our problem statement and historical data, a meticulous preprocessing journey was vital to establish a solid foundation for training the model, and it started with a thorough data cleaning phase. During this initial stage, the data underwent a careful review to detect and eliminate noisy records, and anomalies, including the removal of duplicate records and invalid data points, to ensure the integrity and quality of the dataset. This step was essential to prevent the model from learning from wrong or misleading data, which could have led to biased or inaccurate predictions. Furthermore, the data cleaning phase also involved the handling of missing values, which were either imputed using statistical methods or removed if they were deemed to be irrelevant or redundant. The preprocessing pipeline also included data scaling, which was critical to ensure that all features were on the same scale, preventing features with large ranges from dominating the model's predictions. Additionally, the dataset was also transformed to handle non-linear relationships and interactions between variables, using techniques such as and feature engineering. The outcome of this rigorous preprocessing phase was a high-quality dataset that was well-suited for training a robust and accurate predictive model. The preprocessing journey was a time-consuming process, but it was essential to establish a strong foundation for the model's development, and it ultimately played a critical role in the model's ability to make accurate predictions and provide valuable insights. By carefully preprocessing the data, we were able to unlock the full potential of the predictive model, and provide a reliable and trustworthy solution for predicting the remaining useful life of the choke.

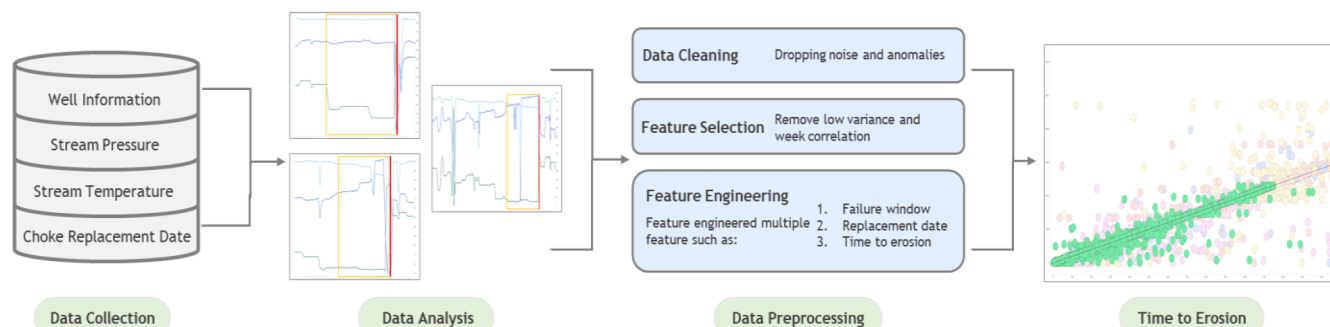


Figure 1—Comprehensive data preprocessing workflow prior to model training.

Predictive Model Description

To build a reliable and scalable solution, a structured machine learning framework was implemented, consisting of all stages from business problem definition all the way to model deployment. The machine learning framework is summarized in Figure 2 and includes the following key stages: task definition, feature

engineering, model evaluation, model selection and finally model deployment. Each stage is crucial to ensuring the development of a robust and accurate model. The framework provides a systematic approach to machine learning, enabling the identification of key factors that impact model performance. By following this framework, we can ensure that our solution is tailored to meet the specific needs of the business, and that it is scalable and reliable.

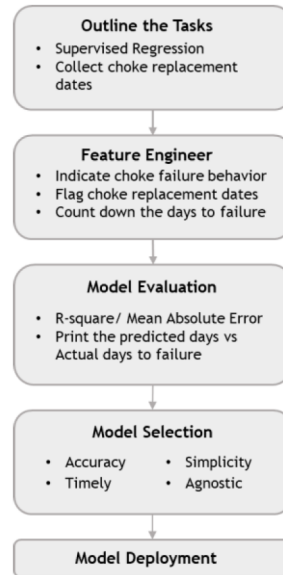


Figure 2—A general ML framework to predict RUL for chokes in gas wells.

Task Definition. The business challenge of determining the remaining days before a choke reaches a critical erosion threshold was formulated as a supervised learning classification problem. The target class represents defined time (days) before erosion, enabling predictive maintenance. This formulation allows for the development of a predictive model that can accurately forecast the remaining useful life of the choke, thereby enabling proactive maintenance scheduling and minimizing the risk of unexpected downtime. By framing the problem in this way, we can leverage the power of machine learning to analyze complex patterns and relationships in the data, and identify the key factors that contribute to choke erosion. The supervised learning approach also enables the use of historical data to train and validate the model, ensuring that it is robust and generalizable to new, unseen data. Furthermore, the regression problem formulation enables the model to provide a clear and actionable output, specifically the number of days before the choke is expected to reach the critical erosion threshold, which can be used to inform maintenance decisions and optimize production operations.

Feature Engineering. In addition to collected real-time data features, a custom target variable was engineered to represent the remaining useful life (days). It is a mathematical decreasing function that begins at the initial recorded timestamp and reaches zero at the erosion date, thereby providing temporal context to the model for estimating days to erosion. This carefully crafted target variable plays a crucial role in enabling the model to learn the complex relationships between the input features and the remaining useful life of the choke. By incorporating this temporal context, the model can effectively capture the dynamic behavior of the system and provide accurate predictions of the remaining days to erosion.

The target is defined as:

$$Target_i = \max(0, (t_{erosion} - t_i))$$

Where:

- t_i = *timestamp of observation i*
- $t_{erosion}$ = *time stamp of erosion event*

Furthermore, the use of a mathematical decreasing function ensures that the model is incentivized to predict earlier erosion dates as the remaining useful life decreases, which is a critical aspect of predictive maintenance. By combining this custom target variable with the collected real-time data features, the model can capture a complex and subtle representation of the choke behavior, which enables it to make accurate and reliable predictions of the remaining useful life. This, in turn, allows for proactive maintenance scheduling and minimizes the risk of unexpected downtime, ultimately leading to improved operational efficiency and reduced costs. The custom target variable is a key component of the predictive maintenance framework, and its careful design and engineering have a direct impact on the accuracy and effectiveness of the model's predictions.

Results and Discussion

The predictive model developed in this study yielded outcomes that presents tangible benefits with an accuracy exceeding 95%. Furthermore, it offers advisory insights to assist production engineers with decision-making processes. The results obtained from the model were seamlessly integrated into a comprehensive advisory dashboard, which enables informed planning, ultimately enhancing the overall efficiency and effectiveness of integrity choke operations in HPHT fields.

The model results are visualized using a dashboard illustrating the RUL of the choke valve for all five wells based on current readings. The days then are colored based on the remaining days to erosion where red is the closest and green represents that the choke is healthy and there is no significant sign of erosion.

This dashboard enables production engineers to gain a deeper understanding of the choke erosion risks by providing a clear view of individual RUL of a choke per well. This contextualized approach empowers engineers to make more informed decisions, especially in HPHT gas field were predicting erosion signs as early as possible allows for a safer environment and improve production efficiency with increased accuracy and reduced downtime always effectively.

The generated insights are currently being utilized as an advisory capacity, with the tool serving as a decision-support system that highlights wells with the nearest predicted vulnerable choke, thus informing prioritization. However, the ultimate decision-making remains with the production engineers, who can leverage their specialized expertise and field experience to review and validate the prediction, ensuring that final decisions are well-informed and contextually relevant.

Overall, the model proves to be a valuable addition to the ongoing choke valve monitoring initiatives, providing engineers with a boosted level of understanding that augmented existing protocols and offered a systematic approach to evaluating erosion risk in conjunction with other operational standers. By integrating the model's insights into their decision-making processes, engineers will be able to make more comprehensive and informed assessments, ultimately strengthening the overall well production management framework.

Conclusion

Harnessing machine learning techniques enables the optimization of well production strategies, proactively minimizing downtime, maximizing equipment efficiency, improving production, avoiding costs, and enhancing overall safety in HPHT gas wells. The AI-driven approach demonstrates strong evidence of effectiveness in predicting choke erosion via the choke Remaining Useful Life (RUL), with an accuracy exceeding 95% in predicting the RUL for gas wells. The trustworthiness of such ML models is crucial for application; therefore, ensuring high historical data reliability, accuracy, and availability, as well as rigorous

dataset preparation and processing, is essential. This requires specialized knowledge in both data science and production engineering for the development and interpretation of the predictive model. Furthermore, in combination with an established digital dashboard powered by AI-predicted RUL, engineers can identify and prioritize wells susceptible to erosion, thereby enhancing surveillance and ensuring well integrity and production efficiency. In conclusion, the successful implementation of this predictive maintenance approach has the potential to streamline the management of HPHT gas wells, enabling operators to capitalize on data-driven techniques and optimize the production strategies.

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